

Complete syllabus checklist

A clear topic-by-topic progress tracker covering all topics in the AI SL course.

	LEARN		REVIEW		CONFIDENT
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1 Number and algebra		17 lessons		
NO.	LESSON			
1.1	Scientific notation			
1.2.1	Arithmetic sequences and series			
1.2.2	Sigma notation			
1.3	Geometric sequences and series			
1.4.1	Compound interest			
1.4.2	Depreciation			
1.4.3	Inflation			
1.5.1	Laws of exponents			
1.5.2	Introduction to logs			
1.5.3	The number e			
1.6.1	Rounding			
1.6.2	Upper and lower bounds			
1.6.3	Percentage error			
1.7.1	Loans			
1.7.2	Annuities			
1.8.1	System of linear equations			
1.8.2	Polynomial equations			

NO.	LESSON			
2.1.1	Gradient of a line	○	○	○
2.1.2	Equation of a straight line	○	○	○
2.1.3	Parallel and perpendicular lines	○	○	○
2.2.1	Functions introduction	○	○	○
2.2.2	Evaluating functions	○	○	○
2.2.3	Inverse functions	○	○	○
2.3	Sketching functions	○	○	○
2.4.1	Features of graphs	○	○	○
2.4.2	Asymptotes	○	○	○
2.5.1	Linear modelling	○	○	○
2.5.2	Quadratic functions	○	○	○
2.5.3	Quadratic modelling	○	○	○
2.5.4	Exponential modelling	○	○	○
2.5.5	Direct and inverse variation	○	○	○
2.5.6	Cubic modelling	○	○	○
2.5.7	Sinusoidal graphs	○	○	○
2.5.8	Creating a sinusoidal model	○	○	○
2.6	Finding parameters of a model	○	○	○

NO.	LESSON			
3.1.1	Midpoint of a line segment	○	○	○
3.1.2	Distance between two points	○	○	○
3.1.3	Volume and surface area	○	○	○
3.1.4	Trigonometry in 3D	○	○	○
3.2.1	Pythagoras' theorem	○	○	○
3.2.2	SOH CAH TOA	○	○	○
3.2.3	Sine rule	○	○	○
3.2.4	Cosine rule	○	○	○
3.2.5	Area of a triangle	○	○	○
3.3.1	Bearings	○	○	○
3.3.2	Angles of elevation and depression	○	○	○
3.4	Arcs and sectors	○	○	○
3.5	Perpendicular bisectors	○	○	○
3.6.1	Voronoi diagram intro	○	○	○
3.6.2	Voronoi examples	○	○	○

NO.	LESSON			
4.1.1	Discrete vs continuous data	○	○	○
4.1.2	Sampling techniques	○	○	○
4.2.1	Histograms and cumulative frequency curves	○	○	○
4.2.2	Box and whisker diagrams	○	○	○
4.2.3	Interpreting box plots	○	○	○
4.2.4	Outliers	○	○	○
4.3.1	Averages and spread	○	○	○
4.3.2	Frequency tables	○	○	○
4.3.3	Constant changes in data	○	○	○
4.3.4	Variance and standard deviation by hand	○	○	○
4.4	Linear regression and correlation	○	○	○
4.5	Probability formulae	○	○	○
4.6.1	Venn diagrams	○	○	○
4.6.2	Tree diagrams	○	○	○
4.7	Discrete random variables	○	○	○
4.8	Binomial distribution	○	○	○
4.9	Normal distribution	○	○	○
4.10	Spearman's rank correlation coefficient	○	○	○
4.11.1	Chi squared intro	○	○	○
4.11.2	Chi squared test for independence	○	○	○
4.11.3	Chi squared goodness of fit	○	○	○
4.11.4	T-test	○	○	○

NO.	LESSON			
5.1.1	Introduction to differentiation	○	○	○
5.1.2	Introduction to limits	○	○	○
5.2	Increasing and decreasing functions	○	○	○
5.3.1	Differentiation power rule	○	○	○
5.3.2	Derivative at a point	○	○	○
5.4	Tangents and normals	○	○	○
5.5.1	Integration (power rule)	○	○	○
5.5.2	Finding $f(x)$	○	○	○
5.5.3	Area under the curve	○	○	○
5.6	Stationary (max/min) points	○	○	○
5.7	Optimisation	○	○	○
5.8	Trapezoidal rule	○	○	○